

Thapar Institute Of Engineering and Technology - Patiala
Computer Science and Engineering Department

Academic Year: 2022-23 - Semester: Second

Course Name : Object Oriented Programming(JAVA) Course Code: UTA028

Exam : MST Marks: 25 Course Coordinator: Dr. Saif Nalband

Date : 07/03/2023 Time : 3:30 PM - 05:30 PM Duration : 120 mins

Name : _____ Roll No: _____

Instructions : There are five questions in total.
Write answer to each question on fresh page.
All the best.

Q1. What are the pillars of Object Oriented Programming Language in JAVA. [4 marks]

Q2. State the difference between

- a) methods and constructors in JAVA
- b) method overloading and method overriding in JAVA

[2+2 marks]

Q3. Write a JAVA program to find the sum of all even numbers and sum of odd numbers in given array of integers.

eg. for given array: nums = {2, 5, 7, 8, 10, 13};

Sum of even numbers: 20 (2 + 8 + 10)

Sum of odd numbers: 25 (5 + 7 + 13)

[3 marks]

Q4. Write a JAVA program that defines a class "Person" with instance variables name, age, and gender.

- a) Create a default constructor that initializes all variables to their default values.
- b) Create a parameterized constructor that takes the name and age as arguments, and sets the gender to "unknown".
- c) Create a method setGender() that takes a string parameter and sets the gender variable to that value.
- d) Additionally, use *this* keyword in your program.

[2+2+2+1 marks]

Q5. Write a JAVA program for creating a class called "Shape" with

- a) a method called getArea() that returns the area of the shape.
- b) Create two subclasses of Shape called "Rectangle" and "Circle".
- c) Override the getArea() method in each subclass to calculate and return the area of the rectangle and circle respectively.

[2+2+3 marks]